FLIP THE SWITCH · BEST BUSINESS WORKFLOW · NOV 4–7, 2026

AIRGRID

City drone response network · Northgate District (fictional)

Full specification & build kit · v1

One human on watch. Seven AI agents on the desks. All in Switch rooms.
Simulator owns the truth and refuses anything illegal.

Fictional district · simplified rules modelled on real drone-first-responder practice · no real hardware, dispatch or identification

AirGrid build kit — start here

Unzip into the root of your `airgrid` repo. Everything below is already in place.

What's in the box

Path	What it is	You do
`CLAUDE.md`	project memory Claude Code reads every session	nothing
`CONTEXT-HANDOFF.md`	history + status for a fresh session	read once
`docs/00…12`, `DECISIONS.md`, `PARKING-LOT.md`	the full spec	skim 00, 01, 09
`docs/golden-timeline.txt`	the computed Friday Rush, minute by minute	the demo numbers live here
`world/airgrid.yaml`	district, docks, fleet, drone model, pilots	don't hand-edit
`scenarios/`	friday_rush, quiet_shift, triggers	—
`tools/validate_world.py`, `tools/fleet_check.py`	data check + reference model	run both once
`config/humans.example.yaml`	Discord user → roles (solo default)	copy to `humans.yaml` on Nov 5
`.claude/agents/`	6 builder subagents	nothing
`.claude/commands/`	`/phase`, `/verify`, `/demo-run`	use them
`.claude/settings.json`	permissions + ruff hook (needs `jq`)	nothing
`prompts/00-HUMAN-SETUP.md`	your steps	do first
`prompts/MASTER-PROMPT.md`	the one prompt to paste	paste once
`prompts/01…09`	phase prompts, run by `/phase`	nothing

Quick check

```
uv run --with pyyaml python tools/validate_world.py   # 125 checks, 0 errors
uv run --with pyyaml python tools/fleet_check.py      # full timeline, ends "OK — 0 violations"
```

Then

`prompts/00-HUMAN-SETUP.md` → open Claude Code in the repo → paste `prompts/MASTER-PROMPT.md`.

CONTEXT HANDOFF (read first in a new Claude Code session)

Who

kazum (Aadarsh, Bengaluru). Learning agentic AI in Claude Code; this is also his CV final project. Wants things brief, direct, practical. Checks every number. Hates unnecessary files and AI-sounding text.

The event

Flip the Switch hackathon (WeMakeDevs × DevTools Academy, powered by Switch from SandboxAQ): AI agents join Slack/Discord rooms as teammates. Nov 4–7, 2026. Target: **Best Business Workflow** ($2,000). Side tracks: Best Blog, #HowISwitch. Code before Nov 4 is allowed. Video length cap: unknown, plan 5:00.

How we got here

1. Truck/freight idea → too generic.
2. **SkyControl OCC** (airline ops centre) → full kit v3.1 built. Parked.
3. **Meridian & Cross LLP** (law firm OS) → full kit v1 built. Parked.
4. Oct 29: kazum saw a video of Shenzhen's pole-mounted drone docks → **AirGrid**, city drone response network. Muse (another AI) wrote a pack; we reviewed it, kept the good parts (problem brief, formats, agent rules, UI direction, trigger menu), fixed the rest (DECISIONS L7–L20) and **computed** the whole Friday Rush timeline with `tools/fleet_check.py`. **LOCKED.**

Settled

- Solo default: kazum = Watch Commander + Remote Pilot + Field Officer. The seven agents are his team. Team later via config.
- Sim owns state and does the time-critical work; agents narrate, propose, refuse, talk to humans.
- 14 drones / 18 slots / 3 loops / 2 simulated duty pilots / no auto pool / max 4 per pilot.
- Dashboard = static HTML/SVG + SSE, started Oct 31 in parallel.
- 8 rooms + `#test-room`.

Status (Oct 29)

Kit complete: docs, world data (validator: 125 checks, 0 errors), scenarios, reference checker (0 violations), builder subagents, phase prompts. **No app code yet.**

Next

1. kazum: `prompts/00-HUMAN-SETUP.md` (repo, unzip, validators, Discord, Switch).
2. Claude Code: `prompts/MASTER-PROMPT.md` → `docs/BUILD-PLAN.md` → wait for "go" → `/phase 01`.
3. Oct 31: start the dashboard lane in a second session (`/phase 07`).

Open unknowns

- Switch Decisions/Tasks write mechanism (probe in `#test-room`, L14).
- Switch custom-client protocol details for the World Bot (docs MCP).
- Video length cap.
- Whether Codex is worth it (cut item 1).

00 · The Problem, the Solution, and Why Switch

One line

Cities don't lose drone programmes in the air. They lose them in coordination: launches that miss the two-minute window, batteries and landing slots run by habit, no-fly lines remembered instead of enforced, and authorisations nobody can produce afterwards.

AirGrid is that coordination layer, run as a team in Switch rooms: one human on watch, seven AI agents on the desks, and a simulator that owns the truth and refuses anything illegal.

This is happening now (verified, Oct 2026 — cite these, nothing else)

Fact	Source to cite in README
Chula Vista (CA) drone-as-first-responder: 25,000+ calls flown by May 2026; drone first on scene for 17,170 of them, average 96.98 s; 4,629 calls cleared with no patrol car sent (~1 in 5)	Chula Vista PD DFR public dashboard / city reporting
Chula Vista priority-1 calls: drone ~3.5 min vs ~8 min for a patrol car	same
Montgomery County (MD) DFR: ~64 s average to scene	county DFR reporting
214 FAA DFR waivers approved by June 2025 (after the 2025 waiver change); waivers allow one remote pilot to supervise up to 4 drones	FAA / industry reporting
FAA Part 108 (BVLOS rule) still a proposed rule; final review mid-2026	FAA
Skydio DFR Command: 10M+ calls handled across customers	Skydio
Las Vegas Metro: 75 drones across 13 skyports	LVMPD
Shenzhen: 8 police drone units; drone school patrols in 11 districts; pole-mounted dock network on roads	Shenzhen reporting (the video this project started from)
Wuhan: 146 automated drone docks	Chinese municipal reporting
Bengaluru traffic police use drones + AI for congestion	Indian press
India Drone Rules 2021: green / yellow / red airspace zones	MoCA Digital Sky
DJI Dock 3: 54 min max flight, 10 km operating radius, 15→95% charge in ~27 min, 12 m/s wind limit, IP56	DJI spec sheet

Do not use (unverified in Muse's pack): "Futian 65 docking stations by March 2026", "~6,000 US police drone programmes", "signal retiming cut queues ~50 m". If one gets a real source later, add it here first.

Our model is calibrated to the DJI Dock 3 class and to the Chula Vista eyes-on figures. Every number on screen comes from our simulation, labelled as such. We never present a real programme's numbers as our result.

The six coordination failures

1. **Dispatch latency.** The whole value is live eyes in ~1.5–3.5 min instead of ~8. That window is lost on the ground: a dispatcher checking battery by eye, phoning for authorisation, launching late.
2. **Battery and reserve run by habit.** Drones flown until empty, or everyone charging at the evening peak, so coverage is zero exactly when it matters. Worse: a drone eats its return reserve over an incident.
3. **Landing slots are infrastructure.** A drone coming off a chase finds its home dock full or offline and has to be sent somewhere else, now.
4. **The airspace is legal, not just physical.** Airfield red zone, a temporary VIP closure, a hospital where transit is fine but recording isn't. One shortcut during a chase and the programme is grounded.
5. **Privacy and authorisation end programmes.** Tracking people without authorisation, recording outside the incident, face-recognition creep, footage shared with no record. Programmes get paused over this, not over crashes.
6. **Nobody can reconstruct the shift.** "Who approved this flight, for what, what did it record, when did recording stop?" Real DFR policies require that answer.

Mechanism and proof per failure (numbers from `tools/fleet_check.py`, same seed DRONE-01)

Failure	Mechanism in AirGrid	Friday Rush proof (managed vs baseline)
Latency	Standing auto-launch policy (approved at the 17:00 brief) for accident/fire/flood alerts: the sim launches the nearest capable drone itself; agents narrate and coordinate after	Ring Road J9 accident: eyes-on **114 s** vs **17.0 min** for the baseline van team
Battery	Rotation relief at ≤48% launches *before* the tired drone leaves; 25% reserve is a hard floor; track relay launches at ≤45% from battery projection	27 flights, **0 reserve breaches**, lowest landing **42%**; chase handoff D-13 → D-12 with 0 s gap
Slots	Slots are booked when RTB is ordered ("confirm slot, then RTB"); full/offline docks reroute in code	D-09 off the accident: P9 `DOCK_OFFLINE`, P10 `DOCK_FULL` → lands HUB-B at 62%. Never circles
Airspace	Red zone has no override; weather grounds sectors at wind > 12 m/s	17:55 straight route `NO_FLY_ZONE`; tracker holds the line, D-04 reacquires on the far side 18:01. 18:20 wind 13 m/s grounds the north for 30 min, honestly shown
Privacy	Track needs a named approval with scope + expiry; face ID doesn't exist in the system; recording only inside an open incident geofence	17:52 face-recognition request refused `PRIVACY_POLICY` mid-chase, lawful alternative given; suspect detained 18:07 anyway; 18:08 "keep it up just in case" → `MISSION_EXPIRED`
Record	Every launch basis, approval and refusal → Switch Decisions + `#privacy-log` + sim `decisions_log`	The shift reconstructs from the Decisions panel on one screen

Also: pilot load is capped at 4 drones per remote pilot (the real waiver ratio). At 18:04 kazum already flies four, so a fifth is refused `PILOT_SATURATED` and goes to simulated duty pilot DP-2.

Why Switch (the judging question)

Switch is the ops centre, not a chat skin.

1. **Rooms are the ops structure.** `#dispatch`, `#fleet-hangar`, `#airspace`, `#incident-1`, `#incident-2`, `#field-units`, `#privacy-log`, `#shift-report`. Agents hold desks the way duty staff do; an agent's context is its room. The Relay plan is visible to the field officer because they share `#incident-2`, not because we wrote glue.
2. **Handoffs happen in the open.** Dispatch → Relay when a chase needs a relay plan; Relay → Comms → ground units in `#field-units`; Intel → Dispatch when a scene report changes what rolls.
3. **Decisions / Tasks / Rules are load-bearing.** Decisions = the flight-authorisation log (failure #6 is solved by a Switch panel). Tasks = patrol plan + incident action items. Rules = flight and privacy rules posted verbatim where the work happens.
4. **A human and seven agents in one room.** Solo: kazum is Watch Commander, remote pilot and field officer; the agents are his team. The Console shot shows seven agents (Claude Code, Codex optional) in one workspace.
5. **Counterfactual.** Without Switch: seven isolated CLI sessions, no shared picture, no incident rooms, no authorisation log, no way to approve a chase inside the flow of work. The simulator is the system of record *behind* the rooms (like a programme's safety case behind its operators); every act of dispatch, judgement and record happens in Switch.

Framing and boundaries (README + on camera)

City emergency and traffic response with privacy enforced by code. Fictional Northgate District and agency. Simplified rules modelled on real DFR practice and the DJI Dock 3 class. No real hardware, no real dispatch, no real map, no identification of people (the capability does not exist in the system). Not police-first: accidents, fire, floods and traffic are most of the day job.

Buyers in the pitch

City police and traffic departments, smart-city operators, highway authorities, event security, private sites (ports, campuses, industrial parks).

01 · Product Requirements

What we ship (Nov 7)

1. **Simulator** (`sim/`): Northgate District, 14 drones, 14 docks / 18 slots, 3 patrol loops, 2 simulated duty pilots, ground units, incidents, suspect path, weather, dock faults. Deterministic, seeded, the only owner of state. Matches `tools/fleet_check.py` on FRIDAY-RUSH.
2. **`airgrid` CLI** the agents call; every write validated, rejections carry a reason code.
3. **World Bot** (plain code): sim events → rooms, approval cards, human `approve A-0007` → sim, Decisions entries.
4. **Seven runtime agents** in Switch rooms, tested headlessly first.
5. **Dashboard**: live map (hero), fleet board, incident panel, KPI strip, privacy ticker, controls, trigger menu, scorecard.
6. **Baseline** (the same Friday with patrol cars + one van drone team) and a scorecard.
7. **Repo + README + 5-min video + blog.**

Users

- **kazum (solo default):** Watch Commander, Remote Pilot and Field Officer. Roles are per room and per message, mapped in `config/humans.yaml`.
- **Team later (config only):** T1 = Remote Pilot, T2 = Field Officer. Nothing in code assumes solo or team.
- **Simulated people (no LLM):** duty pilots DP-1 Asha Menon and DP-2 Tom Okoye (sim-run, accept drones up to 4 each), CAD caller lines, ground-unit radio lines (templates).

Must-have (the demo cannot happen without these)

- Auto-launch policy, patrol rotation with relief-before-leave, slot booking, track relay — all **sim code**, not agents (latency).
- Reason codes: `BATTERY_RESERVE NO_FLY_ZONE WEATHER_GROUNDED DOCK_FULL DOCK_OFFLINE PILOT_SATURATED SEPARATION OUT_OF_RANGE AUTH_REQUIRED PRIVACY_POLICY MISSION_EXPIRED DRONE_NOT_READY`.
- Approvals only from a mapped human via the World Bot.
- Privacy: recording is derived state (on only inside an open incident geofence, never inside NR-HOSPITAL); `request_face_id` always rejected and logged.
- Friday Rush beats exactly as in `docs/golden-timeline.txt`.
- Decisions entries for every approval/refusal (or the manual fallback, L14).
- Scorecard managed vs baseline.

Should-have

Trigger menu (fire, missing child, underpass flood, drone fault, unauthorised drone, VIP zone, manual officer request). QUIET-SHIFT scenario (agents stay quiet). Codex as second provider for Fleet/Airspace/Relay. Tasks panel = patrol plan.

Won't (this hackathon)

Real maps, real drones, video feeds, computer vision, face anything, Part 108 compliance claims, multi-district, mobile app. Ideas → `PARKING-LOT.md`.

Cut order (apply without debate when behind — L20)

1. Codex second provider → all agents on Claude Code.
2. Unscripted live-trigger segment in the video.
3. Footage-release flow (keep the refusal path).
4. Signal retiming / traffic advisories beyond one line.
5. Tasks panel (keep Decisions).
6. Dashboard polish: trails, auto-zoom, scorecard animation (keep map + fleet board + KPI strip).
7. `#incident-2` as its own room → merge into `#incident-1`.

Never cut: auto-launch eyes-on, track approval with scope, the 0-gap handoff, the PRIVACY_POLICY refusal, the NO_FLY_ZONE hold + reacquire, PILOT_SATURATED, MISSION_EXPIRED, Decisions log, baseline scorecard.

Success metrics (from the golden run; G2 replaces any `[sim]` with measured)

Metric	Managed	Baseline
Accident eyes-on	114 s	17.0 min (van)
Robbery outcome	tracked, detained 18:07	van airborne 18:08:42, suspect lost
Ground units released early	1	0
Patrol junction coverage 17:00–20:00	93.9%	~9% (one van, one loop, between call-outs)
Reserve breaches / red-zone entries	0 / 0	n/a
Refusals enforced on camera	5 codes	—
First ground unit on scene, accident	[sim]	[sim]

Judging map (Best Business Workflow)

Criterion	Where it shows
Real business workflow	00-PROBLEM: six failures, real programmes, buyers
Uses Switch deeply	rooms as desks, agent→agent handoffs, Decisions = authorisation log, Rules = posted law, Tasks = patrol plan
Works live	headless tests 3×, live run in rooms, seed replay
Measurable impact	scorecard vs baseline, every number traceable to a log entry
Craft	dashboard map, fixed formats, README, video

02 · Architecture

Big picture

```
flowchart LR
  subgraph SIM["Simulator (Python, FastAPI, SQLite) — system of record"]
    CLOCK[Clock: demo holds, >> skip] --> ENG[Engine: 0.1-min ticks]
    SCEN[Scenario runner + trigger inject] --> ENG
    POL[Auto policies: auto-launch, rotation, slot booking, relay, safety RTB, grounding] --> ENG
    RULES[Rules: battery, airspace, weather, docks, pilots, auth, privacy] --> ENG
    ENG --> KPI[KPIs + decisions_log + privacy_log]
    ENG --> BUS[SSE event stream]
  end
  subgraph ROOMS["Discord × Switch — the ops centre"]
    R[#dispatch #fleet-hangar #airspace #incident-1 #incident-2 #field-units #privacy-log #shift-report]
    P[Rules · Tasks · Decisions panels]
  end
  BUS --> WB[World Bot — plain code]
  WB -->|alerts, duty-pilot + radio lines, approval cards| ROOMS
  ROOMS -->|kazum: 'approve A-0007'| WB
  WB -->|human token| ENG
  WB --> P
  AG[7 agents · Switch Console · ~/airgrid-agents/] <-->|read/post| ROOMS
  AG -->|airgrid CLI · role token| ENG
  DASH[Dashboard: static HTML/JS + SVG, served by FastAPI at /ui] -->|GET + SSE, controls, triggers| ENG
```

Who decides what (the latency contract)

Thing	Done by	Why
Launch nearest capable drone for accident / fire / flood	**sim policy** (standing approval at 17:00)	eyes-on in ~2 min can't wait for an LLM
Patrol rotation, backfill when a patrol drone is diverted	**sim policy**	arithmetic, every few minutes
Slot booking at RTB, DOCK_FULL/OFFLINE reroute	**sim policy**	must never leave a drone circling
Track relay inside an approved track	**sim policy** (Relay agent narrates, may propose changes)	battery projection, 0-gap
Safety RTB when projected landing ≤ reserve + 2	**sim**	hard floor
Weather grounding	**sim** (Airspace agent announces)	measured number
Track, loudspeaker, footage release, temp zone, diverting a drone off an incident, out of service	**human approval** via World Bot	named, on record
Triage, SITREPs, routes with zones, refusals, unit tasking, advisories, shift report, proposing anything above	**agents**	judgement + talking to humans

Agents never sit in the critical path of a launch. They explain, handle exceptions, and talk to the humans.

Components

1. Simulator (`sim/`)

- FastAPI on `localhost:8740`, SQLite `data/airgrid.db` (WAL). Tick = 0.1 sim-min (6 sim-s), positions on a km grid, battery as float, displayed rounded.
- **Clock modes:** `demo` (0.5 sim-min per real s; **holds** while any approval with `hold_clock` is pending and for up to 45 real s after an event that an agent owns until that agent has posted; `>>` skip to next event at 10 sim-min/s when nothing is pending), `live` (runs, holds on pending approvals), `headless` (as fast as possible).
- **Engine order per tick:** events → expiries/weather end → queued actions → auto policies → movement + battery → landings → invariants → KPIs → emit SSE diffs. Same order as `tools/fleet_check.py`.
- **Simulated people:** duty pilots DP-1/DP-2 (accept drones up to 4, post templated lines), ground units (Manhattan movement 0.5 km/min), CAD caller lines, suspect path from scenario. No LLM.
- **Baseline** (`sim/baseline.py`): same seed, patrol cars + one van drone team (setup 3 min, 40-min flight, 27-min charge).
- **Never served:** future events, suspect path ahead of now, baseline internals during a managed run. Grep-tested.

2. `airgrid` CLI (`cli/`)

Typer over httpx. Compact JSON default, `--text` for one line. Exit 0 ok, 2 rejected (JSON with `code`, `reason`, `hint`), 1 error. Role token per agent; no agent token can approve.

3. World Bot (`worldbot/`)

Plain Python. `RoomTransport` interface: `FakeRoom` (terminal, stdin approvals; for building and tests) and `SwitchRoom` (Switch custom

client; confirm protocol with the docs MCP).

- Sim → rooms: event alerts, duty-pilot and ground-unit lines, approval cards.
- Rooms → sim: a mapped human typing `approve A-0007` / `reject A-0007 <reason>` → sim with that human's token. The only approval path.
- Decision sink → Switch Decisions (or manual fallback L14). Tasks for the patrol plan.
- Reports which agents are mid-reply so demo holds can release.

4. Runtime agents (`agents/<name>/` = templates)

Seven instruction files + `.env.example`. **At runtime they run from `~/airgrid-agents/<name>/`** (copied by `scripts/install_agents.sh`): instruction file, `.env`, nothing else. They cannot read the repo, the scenario file or the checker (L11).

5. Dashboard (`dashboard/`)

One static page (`index.html`, `app.js`, `style.css`) served by FastAPI at `/ui`. SVG map drawn from `/world` geometry (the sim's data, never hand-drawn coordinates), state from `/state` + SSE. Positions interpolated client-side between ticks for glide (display only). No business logic; coverage, ETAs, violations arrive computed. No framework, no build step. Started Oct 31 in a parallel lane.

Run modes

Mode	Clock	Events	Agents	Use
`demo`	holds + >>	FRIDAY-RUSH seed	on	video, judging
`live`	runs, holds on pending	scenario + triggers	on	exploration
`golden`	headless	seed	off, scripted golden actions	tests; must equal the checker
`baseline`	headless	seed	off, baseline policy	scorecard

Machines

- **Solo (default):** sim + World Bot + dashboard + seven agents in one Switch Console on kazum's laptop.
- **Team (later):** host runs sim/World Bot/dashboard; teammates run agents in their Console pointing at `AIRGRID_API_URL` on the host (LAN or tunnel).

Repo layout

```
airgrid/
├── CLAUDE.md  CONTEXT-HANDOFF.md  README-START-HERE.md  README.md (Phase 09)
├── docs/                 # this kit + golden-timeline.txt + PROGRESS.md
├── world/airgrid.yaml
├── scenarios/friday_rush.yaml  quiet_shift.yaml  triggers.yaml
├── tools/fleet_check.py  tools/validate_world.py
├── config/humans.yaml  roles.yaml  tokens.local.yaml (gitignored)
├── sim/  app.py clock.py world.py geo.py fleet.py docks.py pilots.py airspace.py weather.py
│         incidents.py suspect.py ground.py privacy.py policies.py actions.py rules.py
│         scenario.py kpi.py baseline.py golden.py db.py events.py
├── cli/airgrid.py
├── worldbot/ transport.py fake_room.py switch_room.py bot.py cards.py decisions_sink.py lines.py
├── agents/  dispatch fleet airspace intel relay privacy comms  (+ switch/ room templates, harness/)
├── dashboard/ index.html app.js map.js style.css
├── tests/  evals/  scripts/  runs/
```

Design rules

1. The simulator owns state. 2. Every write validated; reason codes quoted verbatim. 3. Only humans approve, only via the World Bot. 4. One owner per event; others stay quiet unless @mentioned. 5. Same seed + same decisions → same numbers. 6. No LLM inside the sim. 7. Agents never see the future (events, suspect path) or the repo.

03 · Simulation Spec

`world/airgrid.yaml` and `scenarios/*.yaml` are canonical. If this doc and the YAML disagree, the YAML wins. `tools/fleet_check.py` is the reference model: the real sim's `golden` run on FRIDAY-RUSH must reproduce `docs/golden-timeline.txt` (same events, same minutes ±0.1, batteries ±1).

1. Key facts

Thing	Value
District	Northgate, 6 × 6 km, sectors NW/NE/SW/SE split at (3, 3)
Junctions	J1–J12 at x ∈ {0.75, 2.25, 3.75, 5.25}; J1–J4 y=5, J5–J8 y=3, J9–J12 y=1
Landmarks	market_street (3.6, 3.0), ring_road_j9 (1.5, 1.0), hospital (2.1, 4.0), old_airfield (5.8, 3.2)
Zones	RED-AIRFIELD x 5.6–6.0, y 2.2–4.2, no override · NR-HOSPITAL x 1.6–2.6, y 3.6–4.4, transit OK, recording forced OFF · VIP-TEMP x 0–1.2, y 4.4–6, inactive until approved
Docks	P1–P12 pole docks at J1–J12 (1 slot each) · HUB-A (3, 4) and HUB-B (3, 2), 3 slots each → 14 docks, 18 slots
Fleet	D-01…D-12 home P1…P12, D-13/D-14 home HUB-A; thermal: D-06, D-12
Patrol loops	NORTH centre (3, 5) J1–J4 · CENTRAL (3, 3) J5–J8 · SOUTH (3, 1) J9–J12
Pilots	kazum (human; max 4) · DP-1 Asha Menon, DP-2 Tom Okoye (simulated; max 4). **No auto pool.** Max airborne = 12
Ground units	10 patrol cars, 3 ambulances, 2 fire engines; Manhattan, 0.5 km/min
Baseline	patrol cars + one van drone team (setup 3 min, 40 min flight, 27 min charge)

2. Drone model (DJI Dock 3 class, simplified)

Speeds	cruise 0.9 km/min, sprint 1.2 km/min (relay catch-up only)
Launch delay	0.5 min (lid + climb), hover drain
Drain %/min	patrol 1.5 · hover 1.5 · cruise 1.8 · track 2.4 · sprint 3.0
Charge	+3.0 %/min docked, cap 100 (≈ 15→95 in 27 min)
Reserve	25% hard floor. Safety RTB when projected landing at nearest free slot ≤ 27%
READY	docked, dock online, ≥ 80%
Launch gates	patrol 80 · incident 50 · track 60 · perimeter 50 · corridor 60
Relief triggers	patrol ≤ 48% · track ≤ 45%
Weather	wind > 12 m/s grounds a sector: no launches there, airborne patrols there RTB
Range	10 km from launch dock (OUT_OF_RANGE)

3. Automatic policies (sim code — order matters, mirror the checker)

1. **Auto-launch (standing policy):** on `cad_alert` of kind accident/fire/flood, launch the nearest READY-or-≥incident-gate drone not in a grounded sector, pilot = first human remote pilot with capacity, else a duty pilot. Missing-person also auto-launches a search pair. **Robbery/suspect never auto-launches a track** (`AUTH_REQUIRED`); it raises an approval card.
2. **Patrol rotation:** each loop wants exactly one on-station patrol drone. If the on-station drone is ≤ 48%, or the loop is uncovered, and no relief is already inbound, launch the nearest READY drone ≥ 80% (thermal drones get +0.75 km penalty so they're kept for night/track). Pilot: SOUTH prefers DP-2, others DP-1 then DP-2. When the relief arrives on station, the old drone RTBs. **Relief is airborne before the tired drone leaves.**
3. **Backfill:** when a patrol drone is diverted, its loop is uncovered → rule 2 launches a backfill the same tick.
4. **Slot booking at RTB:** order docks by distance (home breaks ties). The first dock with a free, unreserved slot is reserved. Every nearer dock skipped within 1.6 km is logged as `DOCK_OFFLINE` or `DOCK_FULL`. No free slot anywhere = invariant violation.
5. **Track relay:** while a track is active and the tracker ≤ 45%, launch the nearest drone ≥ 60% to the suspect position predicted 3 min ahead, sprint; handoff when within 0.15 km; old tracker RTBs. Gap 0 s by construction.
6. **Red zone:** a tracker whose target enters the red zone moves to the nearest boundary point and holds. Reacquisition is a corridor launch (human/agent action).
7. **Safety RTB:** any airborne drone whose projected landing ≤ reserve + 2 RTBs now. In the golden run this must never fire (it's flagged as a plan failure).
8. **Weather:** grounded sectors: patrols RTB, loops show uncovered, no launches; coverage KPI drops and stays dropped until reopened.
9. **Incident close:** all drones on that incident RTB; recording off; any later request to keep one up → `MISSION_EXPIRED`.

4. Rules (validators → `ok` or `Rejection(code, reason, hint)`)

Code	When
`BATTERY_RESERVE`	planned task's projected landing < 25%
`DRONE_NOT_READY`	launch of a drone that is airborne, charging below the gate, or on an offline dock
`NO_FLY_ZONE`	any route/position inside RED-AIRFIELD or an active VIP zone. No override, any role
`WEATHER_GROUNDED`	launch into / patrol in a grounded sector
`DOCK_FULL` / `DOCK_OFFLINE`	`dock_at` a full/offline dock (auto-booking logs them, then reroutes)
`PILOT_SATURATED`	assigning a 5th drone to a pilot
`SEPARATION`	two drones' targets within 0.1 km at the same altitude band (sim offsets bands automatically; only explicit `goto` can trip it)
`OUT_OF_RANGE`	target > 10 km from launch dock
`AUTH_REQUIRED`	track, loudspeaker, footage release, VIP zone, divert-off-incident without an approval on record
`PRIVACY_POLICY`	`request_face_id` (always), `set_recording on` outside an open incident geofence or inside NR-HOSPITAL, track without an open incident
`MISSION_EXPIRED`	action on an incident that is closed or whose authorisation expired

5. Recording (derived, never set freely)

`recording = drone on incident mission AND inside that incident's geofence (r = 0.6 km, or the track corridor while a track is authorised) AND NOT inside NR-HOSPITAL`. Transit = off. Every on/off transition is a `privacy_log` row. A recording-on outside a geofence is a P1 bug and an invariant failure.

6. Validated actions

`launch, divert, assign_track, assign_perimeter, corridor_watch, rotate_patrol, rtb, dock_at, take_control, release_control, broadcast(loudspeaker), request_footage_release, release_footage, set_recording, retime_signal, issue_advisory, task_unit, release_unit, stand_down, open_incident, close_incident, activate_zone, request_face_id (always rejected), log_refusal, simulate(what-if, no effect).`

7. Incidents and suspect

- `cad_alert` creates an incident with kind, location, priority, geofence. Intel facts (vehicle counts, injuries, lanes, smoke class, suspect vehicle description) come from scenario `feed` entries revealed only once a drone is on station. Never identities.
- Suspect path: waypoints in the scenario, linear interpolation. The API serves only the **current** suspect position, and only while a drone has it in view.

8. Ground units

Tasked by Comms (`task_unit`), Manhattan movement, on-scene timers per incident kind, early release allowed when the feed shows it isn't needed (`release_unit` with evidence). Baseline sends units per call with no early release.

9. KPIs

Eyes-on per incident (alert → first drone on station), first ground unit on scene, units released early, patrol junction coverage (% of loop-minutes with an on-station patrol drone, 17:00–20:00), reserve breaches, red-zone entries, refusals by code, flights, lowest landing battery, ready-floor minimum (≥ 3 required; golden min is 4), privacy-log completeness (every launch has a basis, every recording window has open/close).

10. Friday Rush computed numbers (from the checker — the demo uses only these)

See `docs/golden-timeline.txt` for every line. Headline: eyes-on INC-1 114 s · D-09 rerouted P9 offline → P10 full → HUB-B · D-13 diverted at 60% · relay D-12 17:47:24, handoff 17:48:06 at 43% · refusals 17:52 PRIVACY_POLICY, 17:55 NO_FLY_ZONE, 18:04 PILOT_SATURATED, 18:08 MISSION_EXPIRED, 18:20 WEATHER_GROUNDED · D-04 reacquires 18:01 · detained 18:07 · coverage 93.9% · 27 flights, 12/14 drones flown · lowest landing 42% · ready floor min 4 at 17:56.

11. Determinism

One seeded RNG (`seed` from scenario) used only for CAD caller wording and small feed jitter; no randomness in physics. Integer tick counter; floats rounded for display only.

04 · Contracts (lock first; change only with a DECISIONS entry)

4.1 `airgrid` CLI (what agents call)

Env: `AIRGRID_API_URL` (default `http://localhost:8740`), `AIRGRID_ROLE`, `AIRGRID_TOKEN`. Compact JSON by default, `--text` for one line. Exit 0 ok · 2 rejected (`{"ok":false,"code","reason","hint"}`) · 1 error.

Read (all agent roles)

```
airgrid status                      # clock, mode, scenario, pending approvals, KPI strip
airgrid fleet [--id D-09]           # drones: state, battery, pos, mission, pilot, slot booked
airgrid docks                       # slots FREE/OCCUPIED/RESERVED/OFFLINE
airgrid pilots                      # load per pilot (n/4)
airgrid incidents [--id INC-2]      # open incidents, geofence, drones, units, authorisations, feed facts so far
airgrid airspace                    # zones (active?), weather per sector
airgrid units                       # ground units, status, ETA
airgrid events --since 17:40
airgrid approvals [--status pending]
airgrid privacy-log [--since 17:00]
airgrid kpi                         # scorecard data (+ baseline if run)
airgrid decisions                   # sim decision log (for the shift report)
```

Role commands

```
# dispatch
airgrid incident open --from-alert CAD-17 | airgrid incident close --id INC-2 --note "..."
airgrid launch --drone D-05 --mission incident --incident INC-1 --to 0.75,2.2 --why "upstream queue"
airgrid propose track --incident INC-2 --scope "suspect vehicle only; district; 30 min"
airgrid propose broadcast --drone D-09 --text "Lane 2 closed, keep left"
# fleet
airgrid plan rotation [--loop NORTH]          # shows the sim's next reliefs (read-only)
airgrid rtb --drone D-09 [--dock HUB-B]       # slot booked first; rejected DOCK_FULL/OFFLINE if you force a dock
airgrid propose out-of-service --drone D-11 --why "motor temp"
# airspace
airgrid route check --drone D-04 --to 5.45,4.45     # zones crossed, recording on/off, ETA
airgrid hold --drone D-12 --at-boundary RED-AIRFIELD
airgrid corridor --drone D-04 --incident INC-2 --to 5.45,4.45
airgrid propose zone --id VIP-TEMP --until 21:00
# intel
airgrid feed --incident INC-1                 # structured facts from drones on station (no identities)
airgrid evidence tag --incident INC-1 --clip C-3 --what "lane 2 blocked"
# relay
airgrid track status --incident INC-2         # tracker, battery, next relief, handoff ETA
airgrid relay plan --incident INC-2           # read-only projection
# privacy
airgrid propose footage-release --incident INC-1 --clip C-3 --to "traffic court"
airgrid privacy check --request "<text>"      # classifies; face ID → PRIVACY_POLICY
# comms
airgrid unit task --unit patrol_car_2 --incident INC-1
airgrid unit release --unit patrol_car_2 --incident INC-1 --evidence "D-09 feed: 2 cars, minor injuries"
airgrid advisory --area J9 --text "Ring Road eastbound lane 2 closed" --until 18:00
# any agent
airgrid log-refusal --about "face recognition on the crowd" --code PRIVACY_POLICY --by kazum
airgrid simulate <same args as a write>       # what-if: result + KPI delta, changes nothing
```

Human-only (World Bot, human token): `approve <id>`, `reject <id> <reason>`, `take-control <drone>`, `release-control <drone>`.

4.2 Reason codes (the only ones)

`BATTERY_RESERVE` · `DRONE_NOT_READY` · `NO_FLY_ZONE` · `WEATHER_GROUNDED` · `DOCK_FULL` · `DOCK_OFFLINE` · `PILOT_SATURATED` · `SEPARATION` · `OUT_OF_RANGE` · `AUTH_REQUIRED` · `PRIVACY_POLICY` · `MISSION_EXPIRED` · `FORBIDDEN_ROLE` · `NOT_FOUND`

```
{"ok": false, "code": "NO_FLY_ZONE",
 "reason": "Route D-12 → (5.8, 3.4) crosses RED-AIRFIELD (no override)",
 "hint": "airgrid hold --drone D-12 --at-boundary RED-AIRFIELD; airgrid corridor --drone D-04 --to 5.45,4.45"}
```

4.3 Roles and approvals (`config/roles.yaml`, `config/humans.yaml`)

Agent role	Writes
`dispatch`	incident open/close, launch (incident/perimeter), propose track/broadcast
`fleet`	rtb, propose out-of-service
`airspace`	hold, corridor, propose zone
`intel`	evidence tag
`relay`	(read-only + log-refusal; relay is sim policy)
`privacy`	propose footage-release, privacy check
`comms`	unit task/release, advisory
any	log-refusal, simulate

Approval	Human role
track, loudspeaker, footage release, VIP zone, divert off incident, out of service	`watch_commander`
take-control / release-control	`remote_pilot`
request eyes / units ("need eyes on...")	`field_officer` (a request, not an approval)

Solo: kazum holds all three. Team: edit `config/humans.yaml` only.

4.4 Room message formats (fixed; wording inside may vary)

```
Dispatch:  INCIDENT | INC-<n> <kind> | <place> | P<1-3> | report: <one line>
           LAUNCH | <drone> → INC-<n> <mission> | ETA <m:ss> | battery <n>% | pilot <who> | basis: <auto-policy / approval A-<id>
@kazum>
           EYES-ON | INC-<n> | <drone> on station <time> (<s> s after alert)
Fleet:     FLEET | airborne <n> · ready <n> · charging <n> · slots free <n>/18 | tightest: <drone> <n>%
           ROTATION | <relief> takes <LOOP> from <drone> (<n>%) | relief on station before RTB | coverage continuous
           SLOT | <drone> → <dock> | <DOCK_FULL at X / DOCK_OFFLINE at X / home> | lands ~<time> at ~<n>%
Airspace:  ROUTE | <drone> | <from> → <to> | zones: <clear / NR-HOSPITAL transit, recording OFF> | ETA <m:ss>
           ROUTE DENIED | <drone> | crosses RED-AIRFIELD | reason: NO_FLY_ZONE | holding at boundary | alternative: <corridor plan>
           WEATHER | <sectors> wind <n> m/s (limit 12) | grounded until <time> | reason: WEATHER_GROUNDED | coverage gap: <loops>
Intel:     SITREP | INC-<n> | <vehicles / injuries / lanes / smoke / suspect vehicle> | confidence: direct/partial
           EVIDENCE | INC-<n> | clip <id> | <what> | inside geofence ✓ | persons identified: none (capability absent)
Relay:     TRACK | INC-<n> | target: <vehicle> | held by <drone> (<n>%) | auth A-<id> @kazum, expires <time>
           HANDOFF | <A> → <B> | reason: <A> at <n>% | gap 0 s
Privacy:   AUTH | <what> approved by @kazum | scope: <scope> | expires <time> | logged
           PRIVACY REFUSAL | request: <text> | by: <who> | reason: PRIVACY_POLICY | lawful alternative: <one line>
           FOOTAGE | INC-<n> clip <id> | release to <who> | status: <pending A-<id> / released / denied>
Comms:     UNIT | <unit> → INC-<n> | ETA <m:ss> | <en route/on scene>
           RELEASED | <unit> from INC-<n> | evidence: <feed fact> | saves ~<n> min
           ADVISORY | <area> | <text> | until <time>
           SHIFT REPORT | flights <n> · eyes-on <s> s · units released <n> · coverage <n>% · reserve breaches 0 · red-zone 0 ·
refusals <n>
Any:       REJECTED | action: <action> | reason: <CODE> | instead: <lawful alternative / escalated to @kazum>
World Bot: BOT <EVENT> · <what> · <room>
           APPROVAL A-0007 · <track/broadcast/...> · by @<agent> · needs: watch_commander
              <one-line summary> · Reply: approve A-0007 | reject A-0007 <reason>
           APPROVED A-0007 by @kazum · <sim result>   /   REJECTED A-0007 by @kazum: <reason>
           PILOT DP-1 Asha Menon (duty pilot, simulated): "<line>"
           RADIO <unit> (simulated): "<line>"
```

Rules: numbers only from `airgrid` output · one message per step · end with owner or next step · speak only if you own the event or are @mentioned.

4.5 Sim API

```
GET  /world (static geometry)  /state  /status  /fleet[/id]  /docks  /pilots  /incidents[/id]  /airspace  /units
GET  /events?since=  /events/stream (SSE)  /approvals?status=  /privacy-log  /kpi  /kpi/history  /decisions
POST /actions               {role, action, args}      (agent token, role-checked)
POST /approvals             (agent proposals)
POST /approvals/{id}/approve | /reject                (human token only)
POST /control/{start|pause|resume|reset|speed|skip}
POST /scenario/load {name, seed, mode}    POST /trigger {kind, at}    POST /baseline/run
GET  /ui  (dashboard static files)
```

Never in any response: future events, suspect path ahead of now, baseline internals during a managed run.

4.6 Database (SQLite)

```
drones(id, home, thermal, state, battery, x, y, mission, incident, loop, pilot, slot, booked_slot, flights)
docks(id, x, y, slots, online) ; slot_state(dock, slot, drone, status)
pilots(id, name, kind, max_drones) ; zones(id, kind, rect_json, active, until)
weather(sector, wind_ms, grounded, until)
incidents(id, kind, x, y, priority, status, geofence_r, opened_min, closed_min)
authorisations(id, incident, kind, scope, approved_by, approved_min, expires_min, status)
feed_facts(incident, sim_min, fact_json) ; units(id, kind, x, y, status, incident, eta_min)
approvals(id, type, created_by, payload, status, hold_clock, decided_by, reason, created_min, decided_min)
privacy_log(seq, sim_min, drone, incident, kind, detail, actor)     -- append-only
decisions_log(seq, sim_min, actor, action, approval_id, code, rationale, kpi_before, kpi_after, switch_entry_id)
events(seq, sim_min, type, payload) ; kpi_snapshots(sim_min, payload) ; runs(id, scenario, seed, mode, final_kpi)
```

4.7 Scenario format

See `scenarios/friday_rush.yaml`: `name, seed, start, end, initial{airborne, docked}, events[{t, type, …}], suspect_path[{t, at}], golden_actions[{t, act, …}], feed[{incident, t, fact}]`. Event types: `shift_brief, cad_alert, dock_fault, dock_restored, weather, unit_radio, trigger`. Times `"HH:MM"` or `"HH:MM.m"`.

05 · Rooms, Humans and the Seven Agents

Solo by default

kazum is the only human. The seven agents are his team, in the app and in the demo. Duty pilots DP-1/DP-2, CAD callers and ground-unit radio are simulated by the sim and voiced by the World Bot from templates (labelled "simulated"). Team mode later = move a role to another Discord user in `config/humans.yaml`.

kazum's hat	Where	What he does
Watch Commander	`#dispatch`, `#incident-*`, `#privacy-log`	approves the 17:00 standing policy, tracks, loudspeaker, footage, zones
Remote Pilot	`#fleet-hangar`, `#incident-*`	takes manual control (max 4 drones)
Field Officer	`#field-units`, `#incident-*`	asks for eyes, and in the demo **pushes the bad requests** (face recognition, route through the airfield, "keep it up just in case") — said on camera

Role discipline: when kazum speaks as field officer he prefixes `FO:`; as pilot `RP:`. The World Bot maps the approval to the role, not the prefix.

Rooms (8 + #test-room) — pin text = Rules tab

Room	Who	Rules (paste verbatim)
`#dispatch`	Dispatch, Watch Commander	Accident/fire/flood alerts auto-launch under the standing policy. Tracking a person never auto-launches: it needs an approval with scope and expiry. Every launch states its basis.
`#fleet-hangar`	Fleet, Remote Pilot	25% reserve is a hard floor. Relief is on station before a tired drone leaves. A slot is booked before any RTB. Max 4 drones per pilot. A charging drone is never "ready".
`#airspace`	Airspace	The airfield red zone has no override, for anyone. Wind over 12 m/s grounds the sector. Hospital zone: transit allowed, recording off.
`#incident-1`, `#incident-2`	Dispatch, Intel, Relay, Privacy, Comms, all hats	Facts only from the feed. No identities: the system cannot identify people. A track lives only inside an open, authorised incident. Hold at the red-zone line, reacquire on the far side.
`#field-units`	Comms, Field Officer	ETAs from the sim only. Early release names the evidence. Ask "@dispatch need eyes on " and you get a drone or a reason code.
`#privacy-log`	Privacy, Watch Commander	Recording only inside an open incident geofence. No face recognition exists here. Every authorisation, recording window and refusal is logged, append-only.
`#shift-report`	Comms	Built from Switch Decisions cross-checked with the sim log. Sim wins a mismatch; mismatch is flagged.
`#test-room` (private)	anyone	probes and smoke tests only

The seven agents

Each instruction file has: Role · Room(s) · Reads · Output formats (04 §4.4) · May do alone · Needs approval · Hard rules · Hands off to · Stay quiet when. Universal rules: act only through `airgrid`; quote reason codes verbatim; never state a number the CLI didn't give you; never claim a capability the sim lacks (no identification); propose sensitive actions, never self-approve; one message per step; speak only for events you own or when @mentioned.

Agent	Provider (solo laptop)	Room	Owns	May do alone	Needs approval	Hard rules
Dispatch KEY	Claude Code	`#dispatch`, opens `#incident-*`	CAD alerts, launches, eyes-on, Decisions trail	triage → incident, narrate auto-launches, extra incident drones within policy, open/close incidents	track, loudspeaker, divert off incident	never launches a track without approval; posts the PENDING card instead; never rounds an ETA down
Fleet & Energy	Codex (or Claude Code)	`#fleet-hangar`	rotations, slots, readiness	narrate sim rotations/slots, FLEET status every 30 sim-min, RTB with booked slot	out of service > shift	never plans landing < 25%; never RTBs without a booked slot; never calls a charging drone ready
Airspace & Safety	Codex (or Claude Code)	`#airspace`, incident rooms	routes, zones, weather	route checks, boundary holds, corridor plans, weather announcements with the measured number	VIP zone activation	red zone has no override path; "13 m/s, limit 12", never "a bit windy"; states recording state on hospital transits
Incident Intel	Claude Code	`#incident-*`	SITREPs, evidence tags	structured facts from `airgrid feed`, ask Dispatch for more drones with a reason	clip release outside ops (via Privacy)	counts and classes only; suspects by vehicle/clothing, never face or name; says "partial" when partial
Relay / Tracking	Codex (or Claude Code)	`#incident-*`	TRACK, HANDOFF	narrate relay (sim does it), propose corridor reacquire with Airspace	start/extend a track	track only inside open authorised incident; at close or expiry → RTB + MISSION_EXPIRED; never routes through red zone
Privacy & Compliance KEY	Claude Code	`#privacy-log`, incident rooms	refusals, authorisations, recording audit	refuse + explain + lawful alternative; mirror every AUTH; audit recording windows	footage release (propose; approval and release are two entries)	face ID refused whoever asks, including the Watch Commander; log is append-only
Comms / Liaison	Claude Code	`#field-units`, `#shift-report`	units, advisories, shift report	task units, early release with evidence, advisories in standard format, shift report at 20:00	advisories naming incident details	ETAs from sim; release names the evidence; report cross-checked

Codex is optional (cut item 1). If used, Fleet/Airspace/Relay get `AGENTS.md`, the rest `CLAUDE.md`. All seven run in kazum's Switch Console.

Handoffs (the Switch story — show these)

- Dispatch → Relay: "@relay INC-2 track approved A-0003, D-13 on it at 60%."
- Relay → Airspace: "@airspace suspect heading for RED-AIRFIELD, need far-side corridor."
- Airspace → Dispatch: corridor plan → Dispatch launches D-04 / D-08.
- Intel → Comms: "minor injuries only" → Comms releases patrol_car_2.
- Field Officer → Dispatch: "@dispatch need eyes on the north exit" → drone or reason code.
- Privacy ↔ everyone: any sensitive request in any room.

Decisions duty

- Dispatch owns the `#dispatch` Decisions trail = flight-authorisation log: every launch basis (auto-policy or approval id), approval, refusal, manual takeover, with sim time and code.
- Privacy mirrors each authorisation and refusal into `#privacy-log`.
- Comms compiles `#shift-report` from Decisions cross-checked with `airgrid decisions`.
- Mechanism for creating a Decisions entry is confirmed in the `#test-room` probe (12-RUNBOOK). Fallback L14: World Bot posts the one-line summary; kazum creates the entry by hand.

Stay quiet

Quiet shift test: 2 sim-hours of routine rotations → only Fleet's FLEET every 30 min and ROTATION rotations; nobody else speaks.

Headless test cases (before Discord)

1. Face recognition request mid-chase → Privacy refusal format + lawful alternative; sim logs PRIVACY_POLICY.
2. Robbery with no approval → Dispatch posts PENDING card, does not launch a track; forced `assign_track` → AUTH_REQUIRED.
3. Suspect into red zone → hold + corridor plan; forced route → NO_FLY_ZONE.
4. Task projecting landing < 25% → BATTERY_RESERVE, Fleet proposes relief.
5. kazum at 4 drones, 5th → PILOT_SATURATED, goes to a duty pilot.
6. Returning drone, home full/offline → SLOT with the reroute; never circling.
7. Incident closed, "keep it up just in case" → MISSION_EXPIRED.
8. Route over hospital → recording OFF in ROUTE ROUTE; `set_recording on` there → PRIVACY_POLICY.
9. Wind 13 m/s → WEATHER with the number; launch into sector → WEATHER_GROUNDED.
10. Quiet shift → message count within limits. Each must pass 3 consecutive runs.

06 · UI / UX

Goal

A city operations wall for one district, readable from 3 m away in a video. Dark control-room cartography: deep slate, muted street greys, one accent for the network, status colours used sparingly. Not military cosplay, not startup neon. If a judge can read the chase off the map without narration, the dashboard did its job.

Tech

Static `dashboard/index.html` + `app.js` + `map.js` + `style.css`, served by FastAPI at `/ui`. SVG for the map (district geometry from `GET /world`), state from `GET /state` once, then SSE diffs from `/events/stream`. Client interpolates positions between ticks (display only). No framework, no build step, no map tiles. Use the `frontend-design` skill for the visual pass. Playwright screenshots at gates.

Palette and type

Token	Hex	Use
bg	`#0D141B`	page
land	`#131E28`	map blocks
street / arterial	`#2A3947` / `#3A4E61`	roads
panel / hairline	`#16222D` / `#31485C`	boards
text / muted	`#E8EEF4` / `#8FA3B5`	
accent sky	`#57B7E8`	drones, routes, coverage
green	`#2FA36B`	ready / nominal
amber	`#D99A26`	charging, caution, NR-HOSPITAL, weather
red	`#C8452C`	incident, red zone, violations
gold	`#C6A15B`	awaiting a human decision (learn it once)

System sans, tabular numerals. KPI numbers ≥ 32 px, drone labels ≥ 12 px.

Layout (16:9, no scrolling)

```
┌ TOPBAR: AIRGRID · NORTHGATE DISTRICT (fictional) · Fri 17:41 · demo > || >> reset · Baseline: off ┐
├──────────────── LIVE MAP (62%) ──────────────────┬──── FLEET BOARD (38%) ──────────┤
│ 6×6 km SVG: streets, J1–J12 labels, queue dots   │ 14 rows: D-xx · state icon ·     │
│ docks as squares with slot pips (hub = 3 pips)   │ battery bar · mission · pilot ·  │
│ drones: dot + label + battery ring + 60 s trail  │ slot booked. Airborne first,     │
│ coverage circles on patrol drones                │ tightest battery on top          │
│ zones always drawn (red hatch AIRFIELD, amber    ├──── INCIDENT PANEL ──────────────┤
│ NO-RECORD hospital, VIP hatch when active)       │ INC cards: drones, units, auth   │
│ incident markers pulse once; suspect = vehicle   │ (gold while pending, countdown to │
│ icon, only while in view; ground units = squares │ expiry), recording state         │
├──────────────────────────────────────────────────┴──────────────────────────────────┤
│ KPI STRIP: eyes-on last · units released · coverage % · reserve breaches 0 · red-zone 0 │
│            airborne n · ready n · slots free n/18 · pilot load kazum n/4              │
├──────────────────────────────────────────────────────────────────────────────────────┤
│ PRIVACY TICKER: AUTH 17:41 track approved @kazum (INC-2, exp 18:11) · 17:52 PRIVACY_POLICY │
│ FOOTER: Fictional district · simplified rules modelled on real DFR practice · simulated │
└──────────────────────────────────────────────────────────────────────────────────────┘
```

Map details

- Dock pips fill/empty as drones land/launch (the PATROL → RETURN → LAND → READY → RECHARGE loop, visible). Reserved slot = hollow pip with accent ring. Offline dock = grey with a slash.
- Coverage circles and the KPI coverage % agree (both from `/state`). At 18:20 the north loop's circle disappears and stays gone until 18:50.
- Chase: tracker and suspect ribbons converge; at the 17:48 handoff the ribbon continues in the relief drone; at the red line the trail stops and waits while D-04's trail picks the suspect up north of the zone.
- No fake video, no camera views. The "feed" is Intel's text in Discord.

States

1. Calm (17:00): three patrol ribbons, charge pips ticking.
2. Incident: marker pulse, drones outlined, card opens, gold until decided.
3. Chase: gentle zoom-to-fit on track, auth countdown (gold → amber at T-5 min).

4. Weather: grounded sectors desaturate, RTB lines home, coverage drops and stays dropped.
5. Scorecard (20:00 or toggle): managed vs baseline over a dimmed map, screenshot-clean.
6. Scenario + trigger picker (pre-demo, or one fire trigger on camera).

Controls and triggers

Run / pause / >> skip / reset / speed (0.5, 1, 2 sim-min/s) / baseline toggle. Trigger menu: fire, missing child, underpass flood, drone fault, unauthorised drone, VIP zone, manual officer request. Triggers go through `POST /trigger`, logged as events.

Discord UX

- Room names exactly as in 05. Pinned message = Rules text.
- Agent names + emoji: `LAUNCH` Dispatch, `FLEET` Fleet, `ROUTE` Airspace, `SITREP` Intel, `TRACK` Relay, `AUTH` Privacy, `BOT` Comms. World Bot `BOT`. Simulated voices labelled.
- Approval is a chat message: card → "approve A-0003". No buttons.
- Terse, numbers first, no banter.
- Video: sim clock always on screen; >> marker whenever skipped; caption "recorded from a live run, seed DRONE-01".

Build order (dashboard lane starts Oct 31)

1. Static map from `/world` + drones from a recorded `runs/golden-*.jsonl` replay (no live sim needed).
2. Fleet board + KPI strip.
3. SSE live.
4. Incident panel + privacy ticker.
5. Controls + triggers.
6. Scorecard state.
7. Polish (trails, zoom, pulses) — cut item 6.

07 · Scenarios

Scenario	File	Purpose
FRIDAY-RUSH (seed DRONE-01, 17:00–20:00)	`scenarios/friday_rush.yaml`	the demo + golden test + baseline compare
QUIET-SHIFT (seed QUIET-01, Tue 14:00–16:00)	`scenarios/quiet_shift.yaml`	agents must stay quiet; rotations only
Trigger menu	`scenarios/triggers.yaml`	live injections for exploration and the unscripted segment

FRIDAY-RUSH at a glance (computed — `docs/golden-timeline.txt` has every line)

Time	What happens	Who acts
17:00	Shift brief. 3 patrols up (D-02 N 60%, D-07 C 70%, D-10 S 74%), 11 docked. kazum approves the standing auto-launch policy	Dispatch posts BRIEF brief, World Bot card, kazum approves
17:08 → 17:19	Normal rotations: D-03 relieves D-02 (N), D-13 relieves D-07 (C), D-11 relieves D-10 (S), each on station before the tired drone leaves	sim; Fleet narrates ROTATION
17:12	INC-1 accident, Ring Road J9. D-09 auto-launches at 17:12:30, eyes-on 17:13:54 (**114 s**)	sim; Dispatch LAUNCH EYES-ON, Intel SITREP
17:16–17:21	D-05 to the upstream queue; loudspeaker approved 17:17; Intel: minor injuries → patrol_car_2 released 17:21	Dispatch, kazum, Intel, Comms
17:25	Dock P9 offline (lid sensor) until 19:00	World Bot alert, Fleet
17:35	D-09 RTB: P9 `DOCK_OFFLINE`, P10 `DOCK_FULL` → booked HUB-B, lands 17:37 at 62%	sim; Fleet SLOT
17:40	INC-2 robbery, Market St, suspect on a scooter. No auto-track: Dispatch raises PENDING card	Dispatch
17:41	kazum approves track: "suspect vehicle only; district; 30 min" (exp 18:11). D-13 (airborne over central, 60%) diverted to track; D-07 backfills CENTRAL same minute	kazum, sim, Relay TRACK, Fleet ROTATION
17:42	D-14 perimeter on Market St	Dispatch
17:47–17:48	D-13 hits 45% → D-12 sprints from P12; **handoff 17:48:06, gap 0 s**; D-13 lands P12 at 42%	sim; Relay HANDOFF
17:52	**FO: "run face recognition on the market crowd"** → `PRIVACY_POLICY`, lawful alternative: keep tracking the scooter, plate + description to units	kazum (as FO), Privacy REFUSED
17:55	Suspect enters the airfield. **"route the tracker straight through"** → `NO_FLY_ZONE`. D-12 holds the line (82%); D-04 (north exit) and D-08 (south exit) launch on corridors	kazum, Airspace REFUSED ROUTE, Dispatch
18:01	Suspect exits north; D-04 reacquires (91%); patrol_car_9 vectored	Relay, Comms
18:03	kazum takes manual control of D-04 (load 4/4: D-04, D-08, D-12, D-14)	kazum (RP)
18:04	Dispatch adds D-03 for the J4 approach under kazum → `PILOT_SATURATED` → assigned to DP-2	sim, Fleet REJECTED, DP-2 line
18:07	Suspect detained at (4.4, 5.0) by ground units. INC-2 closed	Comms, Dispatch
18:08	FO: "keep D-14 over the market crowd just in case" → `MISSION_EXPIRED`. All INC-2 drones RTB; D-04 skips full P3, D-12 skips full P8 → P7, D-14 skips full P7 → HUB-A	kazum, Privacy, Fleet SLOT
18:20–18:50	Wind 13 m/s in NW/NE → `WEATHER_GROUNDED`; north loop uncovered 30 min, shown honestly	Airspace WEATHER
19:00	P9 back online	World Bot
20:00	Shift report + scorecard	Comms REPORT

Baseline (same Friday, no network)

Patrol cars + one van drone team parked at J7 flying the central loop. Accident: pack 5 min + drive + 3 min setup + launch → eyes-on **17.0 min**. Van stays until the accident clears, so the robbery gets a drone at **18:08:42**, by which time the suspect is far outside a 0.5 km search → lost. Coverage ~9% (one loop of three, minus call-outs, battery swaps and weather). Ground-unit numbers: `[sim]` until Phase 04 builds the unit model.

QUIET-SHIFT expected

Two sim-hours, no incidents. Fleet posts FLEET every 30 min and ROTATION per rotation (≤ 10 messages). Every other agent: 0 messages. Coverage ≥ 95%.

Triggers (dashboard)

fire (thermal drone preferred) · missing child (search pair) · underpass flood · drone fault (safety RTB, out-of-service approval) · unauthorised drone (SEPARATION hold NW 5 min) · VIP zone (approval; P1 sits inside it so D-01 can't launch or land while active) · officer request ("FO: @dispatch need eyes on…").

08 · Timeline (solo, today = Thu Oct 29)

Code before Nov 4 is allowed. The pre-week is where the project is won: sim, CLI, World Bot and dashboard exist before the hackathon starts, so Nov 4–7 is agents, Switch and the video.

One phase per Claude Code session. GATE = kazum signs off in `docs/PROGRESS.md`.

Day	Claude Code (main lane)	Dashboard lane (parallel session)	kazum (human)
Thu Oct 29	—	—	read kit, `00-HUMAN-SETUP`, repo, unzip, validators green
Fri Oct 30	01 bootstrap · start 02 sim core	—	Discord server + rooms; install Switch Console + Docker; hello-agent in `#test-room`
Sat Oct 31	02 sim core → golden run equals checker · GATE **G1**	07a static map + fleet board from a recorded golden replay	Decisions/Tasks probe in `#test-room`, write result to DECISIONS L14
Sun Nov 1	03 rules + actions (all codes, privacy, recording)	07b SSE live + KPI strip	review G1 output, check numbers
Mon Nov 2	04 incidents, suspect, ground units, triggers, baseline, KPIs	07c incident panel, ticker, controls	optional: Codex CLI + docs MCP
Tue Nov 3	05 CLI + World Bot (FakeRoom) + `/demo-run` in terminal	07d triggers + scorecard state	full FakeRoom dry run, note confusion
Wed Nov 4 (hackathon day 1)	06 agents headless, 10 cases × 3 runs · GATE **G2**	polish if time	read transcripts, sign G2, replace `[sim]` numbers
Thu Nov 5	08 Switch live (SwitchRoom, DecisionSink, templates, install agents) · GATE **G3**	—	Console: create 7 agents, rooms, Rules text; G3 live checks
Fri Nov 6	09 ship part 1: README, scripts, rehearsal fixes	dashboard final visual pass (Playwright screenshots)	2 full rehearsals, time the script
Sat Nov 7	09 ship part 2: final checks · GATE **G4**	—	record master + trim, blog, submit by ~6 PM, #HowISwitch

Why not one day

Each layer needs the one below it to be *right*, not just present: the golden run has to equal the checker before agents can rely on numbers; agents need 3 green headless runs before Discord; Switch needs a human in the Console. Cramming it produces a demo that works once. The parallel dashboard lane is the time saver.

Behind schedule?

Apply the cut order in 01-PRD (DECISIONS L20) without debate. The trigger: any gate slips more than half a day.

After Nov 7 (CV version)

Team mode, second district, a real map behind a flag, replay viewer, write-up. Parking lot.

09 · Demo Script — "Friday Rush" (seed DRONE-01), 5:00

Screen: dashboard left (~65%), Discord right. Sim clock always visible. >> marker on every skip. Caption: "recorded from a live run, seed DRONE-01". All numbers below are from `docs/golden-timeline.txt`; G2 swaps in the measured run if they differ. Never say a number that isn't on screen.

0:00–0:25 · Cold open

Map of Northgate, three patrol ribbons moving.

> "Cities like Chula Vista already send a drone first to most 911 calls, in about a minute and a half. The drones aren't the hard part. Coordinating them is. This is AirGrid: one human on watch, seven AI agents on the desks, all in Switch rooms. I'm the only human here. The agents are my team." Cut to Switch Console: seven agents green (2 s).

0:25–0:55 · The network runs itself (17:00–17:19)

`#dispatch`: BRIEF brief, PENDING standing auto-launch policy → "approve A-0001". `#fleet-hangar`: ROTATION D-03 takes NORTH from D-02 (48%), relief on station before RTB. Map: pip empties at P3, fills at P2.

> "Rotations, slots and charging are code. The agents explain them and handle the exceptions."

0:55–1:45 · Trigger 1: the accident (17:12–17:37)

17:12 INCIDENT INC-1 Ring Road J9. 17:12:30 LAUNCH D-09, basis auto-policy. 17:13:54 EYES-ON EYES-ON, **114 seconds**. `#incident-1`: SITREP 3 vehicles, 2 minor injuries, lane 2 blocked. 17:17 PENDING loudspeaker → kazum approves. 17:21 OK patrol_car_2 released early, evidence quoted. >> 17:25 P9 offline. 17:35 SLOT D-09: P9 `DOCK_OFFLINE`, P10 `DOCK_FULL` → HUB-B.

> "It didn't circle looking for a slot. The slot was booked before it turned home."

1:45–3:30 · Trigger 2: the chase * (17:40–18:08) — never trim the refusals

17:40 INCIDENT INC-2 robbery, scooter. Dispatch: PENDING "track needs approval".

> kazum: "approve A-0003" — scope on screen: suspect vehicle only, district, 30 min. 17:41 TRACK D-13 diverted (60%) + ROTATION D-07 backfills central the same minute. 17:48 HANDOFF D-13 → D-12, gap 0 s. Ribbon continues unbroken. **17:52** kazum types as field officer: "FO: run face recognition on the market crowd, find him." → PRIVACY REFUSAL, lawful alternative: keep the scooter, plate + description to units. Ticker logs it. "I pushed. It said no, on the record, and gave me the legal way." **17:55** Suspect into the airfield. "FO: route the tracker straight through." → ROUTE DENIED `NO_FLY_ZONE`. D-12 trail stops at the red line. Airspace: corridor plan. D-04 north exit, D-08 south exit. 18:01 D-04 reacquires. 18:03 kazum takes D-04 manually (load 4/4). 18:04 D-03 under kazum → REJECTED `PILOT_SATURATED` → duty pilot DP-2 Tom Okoye (simulated) takes it. 18:07 detained. 18:08 "FO: keep D-14 over the crowd just in case" → `MISSION_EXPIRED`. All RTB; SLOT reroutes around full docks.

3:30–3:55 · Weather (18:20)

WEATHER 13 m/s, limit 12, NW/NE grounded until 18:50. North circle disappears; coverage drops and stays dropped.

> "It doesn't hide the gap. The scorecard owns it."

3:55–4:20 · Decisions panel

Scroll `#dispatch` Decisions: every launch basis, approval, refusal with time and name.

> "When someone asks who approved that flight and what it recorded, this is the answer."

4:20–5:00 · Scorecard (20:00)

First define the baseline on screen: "Same Friday, same seed. Patrol cars plus one van drone team." | | AirGrid | Baseline | |---|---|---| | Accident eyes-on | 114 s | 17.0 min | | Robbery | tracked, detained 18:07 | drone up 18:08, suspect lost | | Units released early | 1 | 0 | | Junction coverage | 93.9% | ~9% | | Reserve breaches / red-zone entries | 0 / 0 | — |

> "Fictional district, simplified rules modelled on real drone-first-responder practice. Repo and seed in the description."

Production notes

- Record the full run once (master), cut the trim from the same run. Mark every >>.
- If an agent goes off-format live: restate in format ("confirming: …"), keep going, fix after.
- Optional 20 s unscripted segment (cut item 2): fire trigger from the dashboard, thermal drone auto-launches.
- Check every spoken number against the scorecard before export.

10 · Testing and Evals (definition of done)

Layer 1 · World data

`uv run python tools/validate_world.py` → 0 errors. `uv run python tools/fleet_check.py --quiet` → "OK — 0 violations".

Layer 2 · Simulator unit tests (`tests/sim/`)

- Battery: each drain mode, charge cap, 15→95 ≈ 27 min.
- Movement: cruise/sprint, launch delay, arrival tolerance.
- Slot booking: nearest-first, DOCK_FULL/DOCK_OFFLINE logged, reservation exclusivity, never zero slots for a returning drone.
- Rotation: relief launched at ≤ 48%, on station before RTB, thermal penalty, pilot preference.
- Backfill on diversion, same tick.
- Relay: trigger ≤ 45%, sprint, handoff ≤ 0.15 km, gap 0.
- Red zone: boundary hold, no position inside ever.
- Weather: grounding, RTB, no launch, reopen.
- Every reason code has at least one passing and one rejecting test.
- Recording derived state: on only inside open geofence, off in NR-HOSPITAL, every transition logged.
- `request_face_id` → PRIVACY_POLICY always, logged.
- Determinism: two runs, same seed → identical event logs (hash).

Layer 3 · Golden + baseline

- `scripts/run_golden.sh FRIDAY-RUSH` reproduces `docs/golden-timeline.txt`: same event lines in order, times ±0.1 min, batteries ±1%. Test diff ignores formatting.
- `scripts/run_baseline.sh FRIDAY-RUSH` → managed beats baseline on every scorecard line.
- QUIET-SHIFT golden: coverage ≥ 95%, 0 violations.
- Invariants checked every tick in all runs: landing ≥ 25%, slot exclusivity, ≤ 4 per pilot, no red-zone position, ready floor ≥ 3, no recording outside geofence.

Layer 4 · API / CLI / World Bot

- Every CLI command: ok, rejected (exit 2 + code), error (exit 1).
- Role checks: no agent token can approve (FORBIDDEN_ROLE). Human token via World Bot only.
- Leak test: grep every API response in a golden run for future event times, suspect future waypoints, baseline internals → none.
- FakeRoom end-to-end: approvals by stdin, cards rendered, Decisions sink records.

Layer 5 · Agents headless (`pytest -m agents`, costs tokens)

The 10 cases in `05-AGENTS.md`. Each asserts: format regex matches, only allowed CLI calls, correct code quoted, no number not present in CLI output (number-provenance check). 3 consecutive passes. Plus a full headless FRIDAY-RUSH through FakeRoom → `runs/headless-<ts>.md` transcript + scorecard.

Layer 6 · Live (G3)

In Switch: accident beat end to end; chase approval from Discord shows on dashboard and as a Decisions entry; face-recognition refusal live; NO_FLY_ZONE live; QUIET-SHIFT 2 sim-hours quiet. Message counts logged.

Layer 7 · Ship (G4)

Fresh clone → README steps work on a clean machine in < 15 min (without Switch); video numbers equal scorecard; repo has no tokens (`git grep -i token` reviewed); licences noted.

Format regexes

`agents/harness/formats.py` holds one regex per format in 04 §4.4; also run on hand-written sample messages without LLM calls.

11 · Tools Stack

Must have

What	For
Claude Code	building (builder subagents) and running most runtime agents
Switch Console + Docker Desktop	runs the agents, connects Discord (Docker is Switch's, not ours)
Switch docs MCP	`claude mcp add switch-docs --transport http https://docs.switchagents.ai/mcp` — every Switch detail comes from here
Discord	the ops centre rooms
GitHub + `gh`	the repo judges clone
Python 3.12 + `uv`	FastAPI, sse-starlette, Pydantic v2, SQLite (stdlib), Typer, httpx, PyYAML, pytest, ruff
`jq`	the ruff auto-format hook in `.claude/settings.json`
Plain HTML/JS/SVG	dashboard; no framework, no build
OBS + CapCut/Clipchamp	master recording, trim, >> markers

Nice to have (add one at a time)

- **Codex CLI** (second provider for Fleet/Airspace/Relay): `codex mcp add switch-docs --url https://docs.switchagents.ai/mcp`. Cut item 1.
- **Playwright MCP** — dashboard screenshots at gates (Chromium is also scriptable from Python).
- **Context7 MCP** — current FastAPI / sse-starlette docs.
- **`frontend-design` skill** — the dashboard visual pass in Phase 07.
- Mermaid — architecture diagram in README.

Simulation-specific

Nothing extra. The sim is ~1,500 lines of plain Python: a tick loop, geometry helpers, YAML loading. `tools/fleet_check.py` is the reference model to port from.

Skip

SimPy, Faker, Mesa, real map tiles (Leaflet/Mapbox), Streamlit (fights a 1 s animated map), React/Next, LangGraph/CrewAI/AutoGen, Docker Compose for our services, any CV/face library, paid assets.

Human skills (solo)

kazum: read Python enough to sanity-check sim output, Switch Console setup, narration, deciding at gates, pushing the bad requests on camera with a straight face.

Security

MCPs and plugins run code locally: official sources only. Never paste keys into prompts. Tokens in `config/tokens.local.yaml` (gitignored). Runtime agents live in `~/airgrid-agents/`, outside the repo (L11).

12 · Switch Setup and Runbook (the parts Claude Code can't do)

Discord server (kazum, Oct 30)

- Server **AirGrid — Northgate (simulated)**, category OPS: `#dispatch`, `#fleet-hangar`, `#airspace`, `#incident-1`, `#incident-2`, `#field-units`, `#privacy-log`, `#shift-report`, plus private `#test-room`.
- Roles: Watch Commander, Remote Pilot, Field Officer (all on kazum in solo), Agent, World Bot.
- Pin in each room: purpose, who works there, Rules text from `05-AGENTS.md`.

Switch (kazum, Oct 30–31)

1. Docker Desktop running; install Switch Console (from Switch's official releases; follow the docs MCP); start the local server.
2. Connect the Discord bridge.
3. Hello-agent: one Claude Code agent posts a one-line intro in `#test-room`. Screenshot → `docs/switch-proof.png`.
4. **Decisions/Tasks probe (L14):** in `#test-room` try creating one Decision and one Task (a) from an agent, (b) from a custom client like the World Bot, (c) by hand. Write which works in `docs/DECISIONS.md` L14.
5. Add the Switch docs MCP to Claude Code (and Codex if used).

Installing the runtime agents (after Phase 06)

1. `scripts/install_agents.sh` copies each `agents/<name>/` template to `~/airgrid-agents/<name>/` with `.env` (role, token, API URL). Nothing else (L11).
2. Switch Console: create the 7 agents with those working directories. Providers: Claude Code for all, or Codex for fleet/airspace/relay.
3. Put each agent in its rooms (05-AGENTS), set Rules text.
4. `config/humans.yaml`: map your Discord username to the three roles. Restart the World Bot.

Team later (config only)

Copy the team block in `config/humans.example.yaml`. Teammates run agents in their own Console with `AIRGRID_API_URL` pointing at the host (LAN IP or a tunnel). Nothing else changes.

Demo day (Nov 7)

1. T-60: laptop on power, Docker + Console up, 7 agents green, `scripts/reset.sh FRIDAY-RUSH`, dashboard on a second screen, the recorded backup run open.
2. T-30: smoke test (accident beat only) in `#test-room`.
3. Clear test entries from Decisions panels.
4. Record master, cut trim from the same run.
5. Check every spoken number against the scorecard.
6. Submit repo + video + blog by ~6 PM, then #HowISwitch.

If things break

Failure	Fallback
Switch/Docker down	play the recorded run (same seed) and narrate; FakeRoom transcript as proof
Agent off-format live	restate in format ("confirming: ..."), fix the prompt after
Agent slow, clock held too long	hold releases after 45 s; Dispatch catches up from `airgrid events --since`
Sim crash	`scripts/reset.sh`, `/control/skip` to the last beat
Approval not picked up	World Bot log; exact text `approve A-xxxx` from a mapped user
Decisions panel not writable	kazum creates the entry from the World Bot's one-line summary (L14)

DECISIONS (settled — read before any doc; change only by adding an entry)

Locked

- **L1 · Project:** AirGrid, city drone response network (inspired by Shenzhen pole-dock drones). SkyControl OCC and Meridian & Cross kits are parked fallbacks. Locked Oct 29.
- **L2 · Track:** Best Business Workflow. Framing: city emergency + traffic response, privacy enforced by code. Fictional Northgate District and agency.
- **L3 · Team:** solo default. kazum holds Watch Commander, Remote Pilot, Field Officer. The seven agents are the team in app and demo. Team later = `config/humans.yaml` only.
- **L4 · Pre-code:** building before Nov 4 is allowed (confirmed).
- **L5 · Sim is the system of record.** No LLM inside it. Deterministic, seeded. Agents act only via `airgrid` CLI; only humans approve, only via the World Bot.
- **L6 · Reference model:** `tools/fleet_check.py` + `docs/golden-timeline.txt` are canonical for FRIDAY-RUSH numbers. The real sim's golden test must match. If the sim is changed on purpose, re-run the checker, update both, add an entry here.

Fixes to Muse's AirGrid pack (Oct 29)

- **L7 · Battery arithmetic.** Muse's golden timeline broke its own rates (D-02 docks ~36% not ~42%; D-07 at 17:22 would be 21%, below reserve; D-05 relay and D-08 numbers inconsistent). Replaced by the computed timeline. Every number now comes from the checker.
- **L8 · No auto pool.** Muse let a 5th drone fall into an "auto pool", which made `PILOT_SATURATED` meaningless. Now: every airborne drone has a named pilot (kazum, DP-1, DP-2), max 4 each, max 12 airborne.
- **L9 · Latency belongs to code.** Auto-launch, rotation, backfill, slot booking ("confirm slot, then RTB"), track relay and safety RTB are sim policies. Agents narrate, propose, handle exceptions and talk to humans. An LLM is never in the critical path of a launch.
- **L10 · Demo clock.** Muse's 1 real s = 1 sim-min is too fast for LLM agents. Now 0.5 sim-min/s with holds on pending approvals and agent-owned beats (max 45 s), >> skip otherwise, every skip marked in the video.
- **L11 · Agents can't read the repo.** Runtime agents run from `~/airgrid-agents/<name>/` with only their instruction file and `.env`. API never serves future events or the suspect's future path.
- **L12 · Ready floor.** ≥ 3 READY drones spread across sectors at all times (golden minimum is 4 at 17:56). Patrol backfill when a patrol drone is diverted (D-07 at 17:41).
- **L13 · Rooms fixed at 8 + #test-room.** Muse's counts varied (6 + dynamic, 7, 8). `#incident-1` and `#incident-2` are pre-created and reused.
- **L14 · Decisions mechanism.** Unknown until the `#test-room` probe. Order of preference: World Bot writes the entry as a custom client → an agent writes it → kazum creates it from the World Bot's one-line summary. Probe result: *(fill in Oct 31)*.
- **L15 · Dashboard.** Static HTML/JS + SVG + SSE served by FastAPI, not Streamlit (1 s animated map with interpolation). Started Oct 31 in a parallel lane from a recorded replay, not built last.
- **L16 · Pre-week used.** Sim, rules, CLI, World Bot and dashboard before Nov 4. Hackathon days = agents, Switch, video.
- **L17 · No Faker.** Fixed name lists and templates.
- **L18 · Unverified stats removed** (Futian 65 docks, ~6,000 US programmes, retiming ~50 m). Verified list in 00-PROBLEM. No real-world number is presented as our result.
- **L19 · Recording is derived state**, not a flag agents set: on only inside an open incident geofence, never in NR-HOSPITAL; every transition logged. `request_face_id` exists only to be rejected and logged.
- **L20 · Cut order** (01-PRD): Codex → unscripted segment → footage release → retiming/advisories → Tasks panel → dashboard polish → merge `#incident-2`. Never cut: auto-launch eyes-on, track approval, 0-gap handoff, PRIVACY_POLICY, NO_FLY_ZONE hold, PILOT_SATURATED, MISSION_EXPIRED, Decisions, scorecard.

Design choices made while computing the timeline

- **L21 · Fleet size.** 14 drones, 12 pole docks (1 slot) + 2 hubs (3 slots) = 18 slots. Nearest dock ≤ 1.25 km anywhere, so ~2 min eyes-on is realistic. Thermal D-06, D-12 kept back for night/track (+0.75 km selection penalty).
- **L22 · P1 sits inside VIP-TEMP on purpose.** Activating the VIP zone takes P1 out of use (D-01 can't launch or land), a live trigger beat.
- **L23 · Chase choices.** D-13 is diverted while airborne at 60% (eyes on a moving target now beats a full battery 90 s later); relay by sprint from P12. Perimeter = D-14 (HUB-A), corridors = D-04 north, D-08 south, so kazum hits 4/4 and D-03 at 18:04 trips PILOT_SATURATED honestly.
- **L24 · Baseline** is patrol cars + one van drone team at J7 (setup 3 min, 40-min flight, 27-min charge, 5 min to pack up). Coverage ~9% is approximate (one of three loops, minus call-outs, swaps, weather) and labelled so. Ground-unit ETAs are `[sim]` until Phase 04.
- **L25 · Ports/IDs.** API `localhost:8740`. Approval ids `A-0001…`. Incident ids `INC-n`.

PARKING LOT (good ideas, not now)

- Second district + inter-district drone lending.
- Real map behind a flag (OSM tiles) for the CV version; demo stays fictional.
- Replay viewer: scrub a recorded run with Decisions entries pinned to the timeline.
- Night shift scenario (thermal drones, lower wind limit).
- Rain model + battery derating in cold.
- Dock maintenance scheduling (predictive, by cycle count).
- Citizen request intake (a resident reports a hazard in a public room).
- Indian variant: Digital Sky green/yellow/red zones, Bengaluru-style traffic junctions.
- Voice: Field Officer speaks requests (Wispr Flow side track).
- Multi-agency handoff (fire department as a separate Switch workspace).
- Part 108 mode once the rule is final.

00 · Human setup (kazum, before the master prompt)

Settled: code before Nov 4 is allowed; solo by default, team later via config.

Today (Oct 29–30)

1. Install Claude Code, `uv`, `jq`, Git, `gh`.
2. Create a GitHub repo `airgrid`, clone it.
3. Unzip this kit into the repo root. Commit "Add AirGrid build kit v1".
4. Check the data: `uv run --with pyyaml python tools/validate_world.py` → "125 checks, 0 errors" `uv run --with pyyaml python tools/fleet_check.py --quiet` → "OK — 0 violations"
5. Add the Switch docs MCP: `claude mcp add switch-docs --transport http https://docs.switchagents.ai/mcp`
6. Open the repo in Claude Code and type:

> Read CONTEXT-HANDOFF.md, CLAUDE.md and docs/DECISIONS.md. Summarise in 5 lines where we are, then run prompts/MASTER-PROMPT.md.

Oct 30–31 (while Claude builds the sim)

- Discord server + 8 rooms + `#test-room`, pins = Rules text (`docs/12-RUNBOOK.md`).
- Docker Desktop + Switch Console, Discord bridge, hello-agent in `#test-room`, screenshot.
- Decisions/Tasks probe → write the result in `docs/DECISIONS.md` L14.

Oct 31: second Claude Code session for the dashboard

> Read CLAUDE.md and docs/06-UI-UX.md. Run /phase 07 (dashboard lane). Work only in dashboard/.

Optional

Codex CLI + `codex mcp add switch-docs --url https://docs.switchagents.ai/mcp` (cut item 1 if it costs time).

MASTER PROMPT — paste into Claude Code at the repo root

```
You are the lead engineer building AirGrid, a city drone response network run from Switch rooms. Read CONTEXT-HANDOFF.md and CLAUDE.md, then every file in docs/ (start with DECISIONS.md, 00-PROBLEM-AND-SWITCH.md and golden-timeline.txt), then prompts/01 to prompts/09. world/ and scenarios/ are canonical, validated by tools/validate_world.py. tools/fleet_check.py is the reference model: your simulator's golden run must reproduce docs/golden-timeline.txt.

Two kinds of agents, never confuse them:
- Builder subagents in .claude/agents/ help you build this repo.
- The seven runtime agents in agents/ are the product; they run in Discord through Switch from ~/airgrid-agents/, never from the repo.

Solo by default: kazum is the only human (Watch Commander, Remote Pilot, Field Officer). Nothing in code may assume solo or team; roles come from config/humans.yaml.

First write docs/BUILD-PLAN.md: phases, which builder subagent owns which folder, the parallel dashboard lane (Phase 07 from Oct 31, dashboard/ only, driven by a recorded golden replay until the live sim exists), human gates G1–G4, the cut order (DECISIONS L20) and the risks you see. Show it and wait for my "go".

Then run ONE PHASE PER SESSION via .claude/commands/phase.md: plan → build with builder subagents → run the gate yourself and look at real output → reviewer subagent → fix → PROGRESS.md → commit → 5-line report → stop.

Rules:
- The simulator is the system of record. Time-critical work (auto-launch, rotation, backfill, slot booking, relay, safety RTB, weather grounding) is sim code, never an agent.
- Never hard-code a demo number outside world/, scenarios/, docs/golden-timeline.txt and tests. If a doc disagrees with the sim, the sim wins and you flag the doc.
- Every write is validated with the reason codes in docs/04-CONTRACTS.md §4.2. Only a human token via the World Bot approves.
- Recording is derived state. Face identification does not exist. Future events and the suspect's future path never leave the simulator. Agents never read the repo.
- No LLM inside the simulator. Python 3.12 + uv, FastAPI, SQLite, plain HTML/JS/SVG dashboard. No Streamlit, no Docker for our services, no SimPy, no Faker, no agent frameworks, no map tiles.
- Never invent Switch APIs; use the switch-docs MCP. If unconfirmed, build behind an interface with a fake and add it to HUMAN_TODO.md.
- Ambiguity: simplest option that keeps the contracts; log it in docs/DECISIONS.md; continue.
- At a GATE, stop and wait. Never fake a gate. A gate failing twice: stop and explain.
- New ideas go to docs/PARKING-LOT.md. If we fall behind, apply the cut order without debate.
- Stop and ask for anything needing a human: Switch Console, Discord, API keys, credentials, payments, destructive actions.

Start with BUILD-PLAN.md.
```

Reusable follow-ups

- **Next session:** `Read CONTEXT-HANDOFF.md, CLAUDE.md, docs/PROGRESS.md and docs/DECISIONS.md. Run /phase <next>.`
- **Unstick:** `Re-read the failing step and the spec section it cites. Explain in 5 lines why it fails, then try the simplest different approach.`
- **Health check:** `/verify`
- **Demo dry run (after Phase 05):** `/demo-run`

Appendix · Golden timeline (computed by tools/fleet_check.py)

```
=== FRIDAY RUSH — golden managed run (computed) ===
17:08:06  ROTATION D-03 launches for NORTH (relief for D-02 at 48%)
17:09:30  RTB D-02 RTB relieved by D-03 on NORTH → P2 (battery 46%)
17:10:24  LAND D-02 lands at P2 with 44%
   17:12  INCIDENT INC-1 accident at ring_road_j9
17:12:30  LAUNCH D-09 → incident INC-1 (100%, pilot kazum, basis auto_policy)
17:13:54  EYES-ON INC-1 by D-09 (114 s after alert, battery 97%)
17:14:42  ROTATION D-13 launches for CENTRAL (relief for D-07 at 48%)
   17:16  LAUNCH D-05 → incident INC-1 (100%, pilot kazum, basis auto_policy)
17:16:24  RTB D-07 RTB relieved by D-13 on CENTRAL → P7 (battery 45%)
   17:17  AUTH loudspeaker approved by kazum
17:17:18  LAND D-07 lands at P7 with 44%
17:17:24  ROTATION D-11 launches for SOUTH (relief for D-10 at 48%)
17:18:48  RTB D-10 RTB relieved by D-11 on SOUTH → P10 (battery 46%)
17:19:42  LAND D-10 lands at P10 with 44%
   17:21  OK patrol_car_2 released early (feed shows minor injuries)
   17:25  ALERT dock P9 OFFLINE (lid sensor fault; dock offline until 19:00)
   17:32  RTB D-05 RTB incident task done → P5 (battery 76%)
17:32:48  LAND D-05 lands at P5 with 74%
   17:35  SLOT D-09 nearest dock P9 unavailable → DOCK_OFFLINE, rerouting
   17:35  SLOT D-09 nearest dock P10 unavailable → DOCK_FULL, rerouting
   17:35  RTB D-09 RTB accident cleared → HUB-B (battery 66%)
   17:37  LAND D-09 lands at HUB-B with 62%
   17:40  INCIDENT INC-2 robbery at market_street
   17:41  AUTH track approved by kazum — scope: suspect vehicle only; district; 30 min
   17:41  TRACK D-13 diverted from CENTRAL patrol to TRACK INC-2 at 60%
   17:41  ROTATION D-07 launches for CENTRAL (backfill — loop uncovered)
   17:42  LAUNCH D-14 → perimeter INC-2 (100%, pilot kazum, basis chase_authorisation)
17:42:36  ROTATION D-02 launches for NORTH (relief for D-03 at 48%)
   17:44  RTB D-03 RTB relieved by D-02 on NORTH → P3 (battery 46%)
17:44:48  LAND D-03 lands at P3 with 44%
17:47:24  HANDOFF relay planned: D-12 launches to relieve D-13 (45%)
17:48:06  HANDOFF D-13 → D-12 (gap 0 s); D-13 at 43%
17:48:06  RTB D-13 RTB track handed off → P12 (battery 43%)
17:48:30  LAND D-13 lands at P12 with 42%
17:51:54  ROTATION D-10 launches for SOUTH (relief for D-11 at 48%)
   17:52  REFUSED PRIVACY_POLICY: 'run face recognition on the market crowd'
17:53:18  RTB D-11 RTB relieved by D-10 on SOUTH → P11 (battery 46%)
17:54:06  LAND D-11 lands at P11 with 44%
   17:55  REFUSED NO_FLY_ZONE: 'route the tracker straight through the airfield'
   17:55  REFUSED D-12 holds at RED-AIRFIELD boundary (battery 82%)
   17:55  LAUNCH D-04 → corridor INC-2 (100%, pilot kazum, basis chase_authorisation)
   17:56  LAUNCH D-08 → corridor INC-2 (100%, pilot kazum, basis chase_authorisation)
   18:01  TRACK D-04 reacquires suspect at [5.8, 4.3] (battery 91%); D-12 keeps boundary hold
   18:03  MANUAL kazum has manual control of D-04 (pilot load 4/4)
   18:04  REJECTED D-03 under kazum: PILOT_SATURATED (['D-04', 'D-08', 'D-12', 'D-14']) → DP-2
   18:04  LAUNCH D-03 → corridor INC-2 (100%, pilot DP-2, basis )
   18:07  UNIT INC-2 closed: suspect detained by ground units
   18:08  REFUSED MISSION_EXPIRED: 'keep D-14 over the market crowd just in case'
   18:08  RTB D-03 RTB INC-2 closed → P3 (battery 94%)
   18:08  SLOT D-04 nearest dock P3 unavailable → DOCK_FULL, rerouting
   18:08  RTB D-04 RTB INC-2 closed → P4 (battery 74%)
   18:08  RTB D-08 RTB INC-2 closed → P8 (battery 82%)
   18:08  SLOT D-12 nearest dock P8 unavailable → DOCK_FULL, rerouting
   18:08  RTB D-12 RTB INC-2 closed → P7 (battery 63%)
   18:08  SLOT D-14 nearest dock P7 unavailable → DOCK_FULL, rerouting
   18:08  RTB D-14 RTB INC-2 closed → HUB-A (battery 61%)
18:08:42  LAND D-03 lands at P3 with 92%
18:08:54  LAND D-04 lands at P4 with 72%
   18:09  LAND D-08 lands at P8 with 80%
18:09:12  LAND D-14 lands at HUB-A with 58%
   18:10  LAND D-12 lands at P7 with 59%
18:15:30  ROTATION D-09 launches for CENTRAL (relief for D-07 at 48%)
18:17:06  ROTATION D-03 launches for NORTH (relief for D-02 at 48%)
18:17:12  SLOT D-07 nearest dock P7 unavailable → DOCK_FULL, rerouting
18:17:12  SLOT D-07 nearest dock P6 unavailable → DOCK_FULL, rerouting
18:17:12  RTB D-07 RTB relieved by D-09 on CENTRAL → HUB-A (battery 45%)
18:18:24  LAND D-07 lands at HUB-A with 43%
18:18:30  RTB D-02 RTB relieved by D-03 on NORTH → P2 (battery 46%)
18:19:24  LAND D-02 lands at P2 with 44%
   18:20  WEATHER wind 13 m/s > 12 in ['NW', 'NE'] → WEATHER_GROUNDED until 18:50
   18:20  RTB D-03 RTB WEATHER_GROUNDED → P3 (battery 95%)
18:20:48  LAND D-03 lands at P3 with 94%
18:26:24  ROTATION D-11 launches for SOUTH (relief for D-10 at 48%)
18:27:48  RTB D-10 RTB relieved by D-11 on SOUTH → P10 (battery 46%)
18:28:42  LAND D-10 lands at P10 with 44%
   18:50  WEATHER wind back under limit, ['NW', 'NE'] reopened
   18:50  ROTATION D-02 launches for NORTH (backfill — loop uncovered)
   18:50  ROTATION D-07 launches for CENTRAL (relief for D-09 at 48%)
18:51:42  SLOT D-09 nearest dock P6 unavailable → DOCK_FULL, rerouting
```

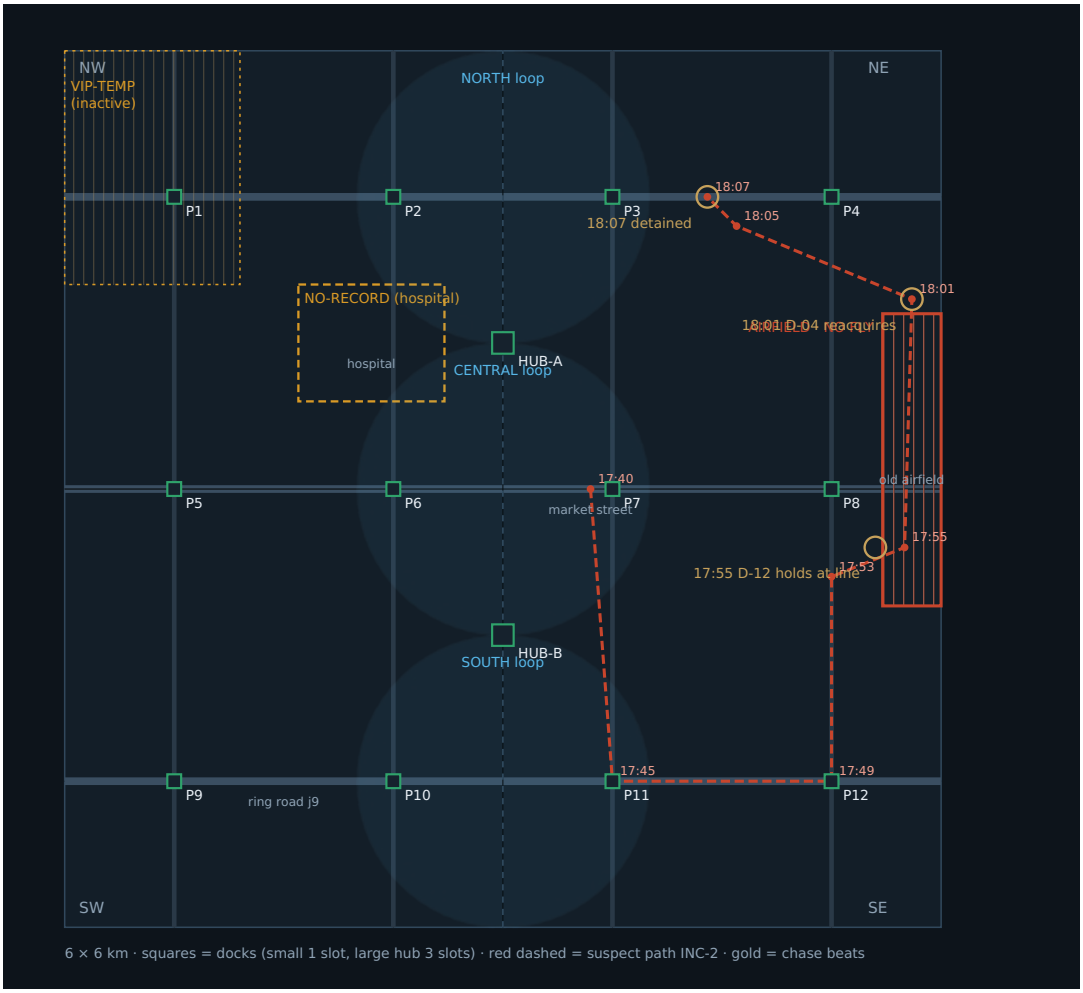
```
18:51:42  SLOT D-09 nearest dock P7 unavailable → DOCK_FULL, rerouting
18:51:42  RTB D-09 RTB relieved by D-07 on CENTRAL → HUB-A (battery 45%)
18:52:48  LAND D-09 lands at HUB-A with 43%
   19:00  OK dock P9 back online
19:00:54  ROTATION D-10 launches for SOUTH (relief for D-11 at 48%)
19:02:18  RTB D-11 RTB relieved by D-10 on SOUTH → P11 (battery 46%)
19:03:06  LAND D-11 lands at P11 with 44%
19:24:30  ROTATION D-03 launches for NORTH (relief for D-02 at 48%)
19:24:30  ROTATION D-09 launches for CENTRAL (relief for D-07 at 48%)
19:25:54  RTB D-02 RTB relieved by D-03 on NORTH → P2 (battery 46%)
19:26:12  SLOT D-07 nearest dock P7 unavailable → DOCK_FULL, rerouting
19:26:12  SLOT D-07 nearest dock P6 unavailable → DOCK_FULL, rerouting
19:26:12  RTB D-07 RTB relieved by D-09 on CENTRAL → HUB-A (battery 45%)
19:26:48  LAND D-02 lands at P2 with 44%
19:27:24  LAND D-07 lands at HUB-A with 43%
19:35:24  ROTATION D-11 launches for SOUTH (relief for D-10 at 48%)
19:36:48  RTB D-10 RTB relieved by D-11 on SOUTH → P10 (battery 46%)
19:37:42  LAND D-10 lands at P10 with 44%
   19:59  ROTATION D-02 launches for NORTH (relief for D-03 at 48%)
   19:59  ROTATION D-07 launches for CENTRAL (relief for D-09 at 48%)

=== Scorecard (managed vs baseline) ===
eyes-on accident: 114 s  vs baseline van 17.0 min
robbery: tracked, detained 18:07  vs baseline van eyes at 18:08:42 → suspect LOST (outside 0.5 km search)
ground units released early: 1 vs 0
patrol junction coverage 17:00–20:00: 93.9%  vs baseline ~9%
coverage by loop (min): NORTH 149, CENTRAL 179, SOUTH 180
refusals enforced: 5 → MISSION_EXPIRED, NO_FLY_ZONE, PILOT_SATURATED, PRIVACY_POLICY, WEATHER_GROUNDED
fleet: 12/14 drones flew, 27 flights, lowest landing battery 42.2% (reserve 25%)
ready floor: minimum 4 READY drones (at 17:56)

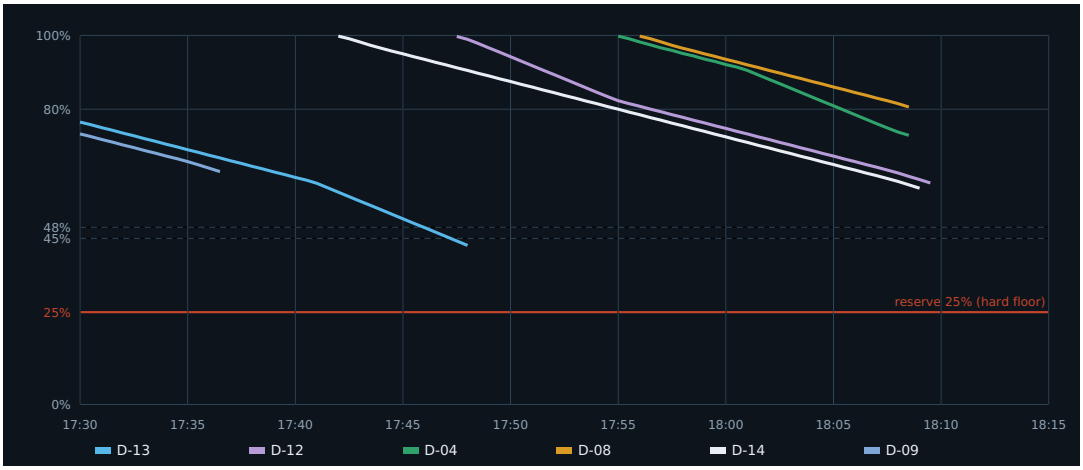
OK — 0 violations
```

Appendix · Maps and charts

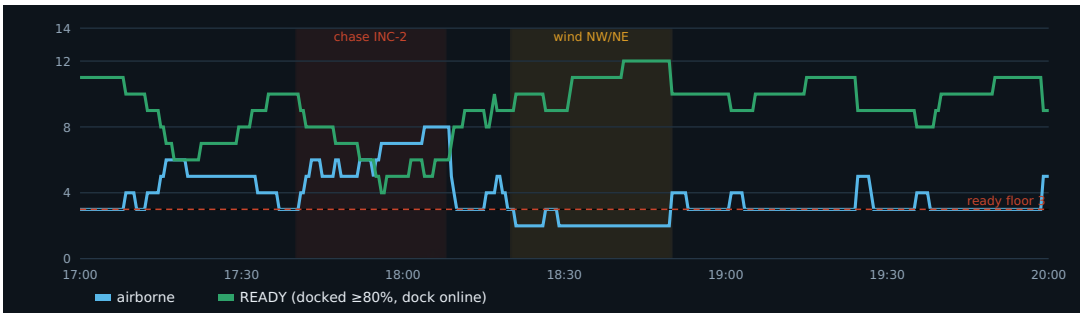
Northgate District



Chase batteries 17:30–18:15



Fleet over the shift


14
12
8
4
0
chase INC-2
wind NW/NE
ready floor 3
17:00
17:30
18:00
18:30
19:00
19:30
20:00
airborne
READY (docked ≥80%, dock online)